Theoretical Foundations of Statistical Distributions

1. What is Statistical Distributions?

Answer:

A **statistical distribution** describes how values of a random variable are spread across possible outcomes. It shows both the **range of values** and the **likelihood (probability)** of each occurring.

- **Key Functions**:
  - PMF → For discrete outcomes (coin toss).
  - PDF → For continuous outcomes (height, weight).
  - CDF → Probability of that variable ≤ a certain value.

📐 Formula (General Probability Distribution)

Probability Distribution Formula

Mean ($\bar{x}$) = $\sum [x_i * P(x_i)]$

Standard Deviation (σ) = $\sqrt{\sum (x_i - \bar{x})^2 * P(x_i)}$

📊 Example

- Tossing a fair coin:
  - Probability of Head = 12
  - Probability of Tail = 12

This is a **Bernoulli distribution** (special case of statistical distribution with two outcomes)

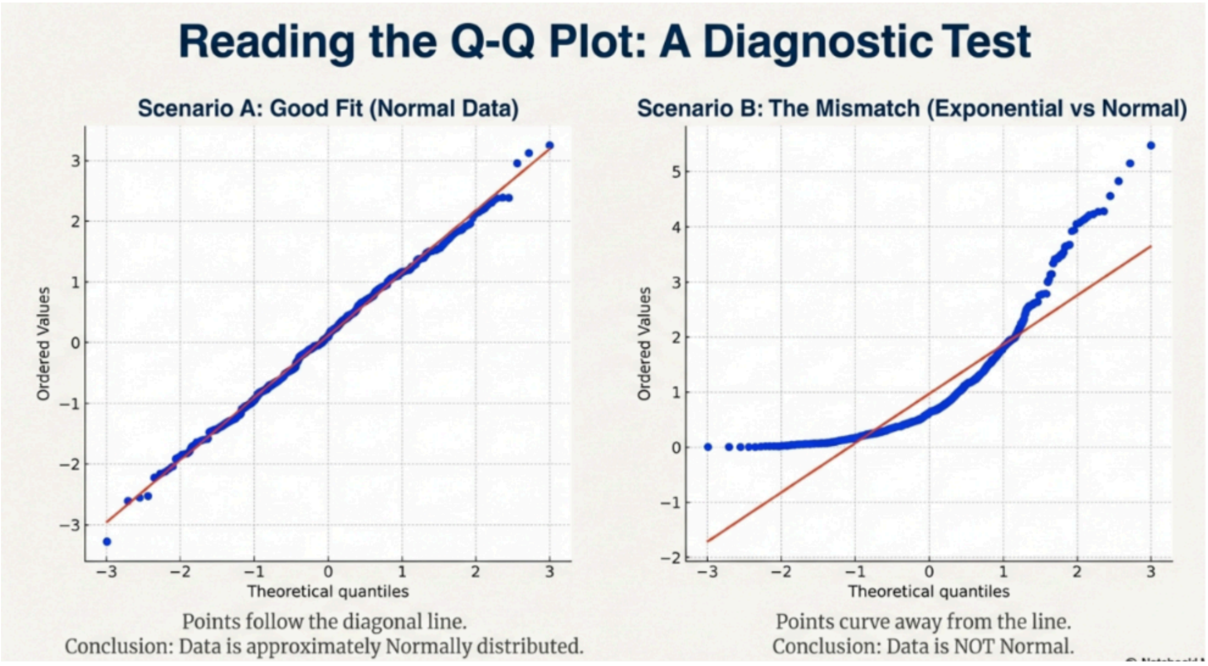
2: What is a Q-Q Plot and why is it used?

Answer:

A **Q-Q Plot (Quantile-Quantile Plot)** is a graphical tool used to compare the quantiles of a dataset against the quantiles of a theoretical distribution (commonly the normal distribution).

- **Purpose**: To check if data follows a specific distribution.
- **Interpretation**:
  - Points on a straight diagonal line → Data fits the distribution.
  - Deviations from the line → Indicate skewness, heavy tails, or mismatch.

📊 Plot



📊Example

Suppose we have exam scores of 100 students.

- If we plot their quantiles against a **normal distribution's quantiles**, and the points align closely with the diagonal line → scores are approximately normally distributed.
- If points curve upward → data is **right-skewed**.
- If points curve downward → data is **left-skewed**.

3: Difference between Discrete and Continuous Distributions

Answer:

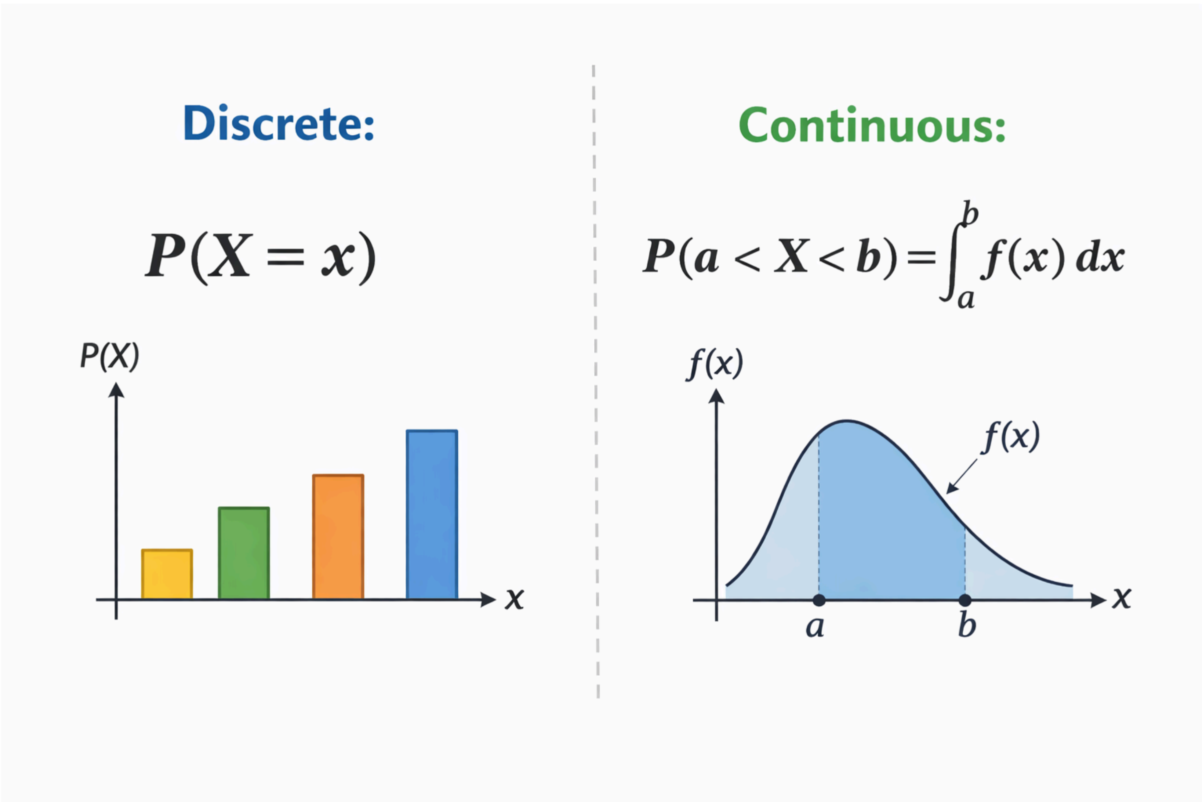
A **distribution** describes how probabilities are assigned to possible outcomes of a random variable.

The key difference lies in the **type of values** the variable can take.

Feature	Discrete Distribution	Continuous Distribution
Definition	Probability for countable	Probability for infinite

	outcomes	outcomes within a range
Examples	Coin toss, number of emails per hour	Height, weight, time
Function Type	PMF (Probability Mass Function)	PDF (Probability Density Function)
Probability of exact value	Non-zero	Almost zero
Graph Type	Bar chart	Smooth curve
Formula	P(X=x)	f(x)=dF(x)dx

📐 Formula Representation



📘Example

- **Discrete**: Rolling a die → outcomes {1, 2, 3, 4, 5, 6}.
- **Continuous**: Measuring height → any value between 150 cm and 200 cm.

4: What is Bernoulli Distribution?

Answer:

The **Bernoulli Distribution** is the simplest type of discrete probability distribution.

It models a **single trial** that has only **two possible outcomes — Success (1)** or **Failure (0)**.

- **Parameter**: p = probability of success
- **Range**: $X \in \{0,1\}$
- **Mean**: $\mu = p$
- **Variance**: $\sigma 2 = p(1-p)$

📐 Formula

BERNOULLI DISTRIBUTION

- A Bernoulli trial is an experiment with only two outcomes. An r.v. X has Bernoulli(p) distribution if

$$X = \begin{cases} 1 \text{ with probability } p \\ 0 \text{ with probability } 1-p \end{cases} ; 0 \le p \le 1$$

$$P(X = x) = p^x (1-p)^{1-x} \text{ for } x = 0,1; \text{ and } 0 < p < 1$$

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📊Example

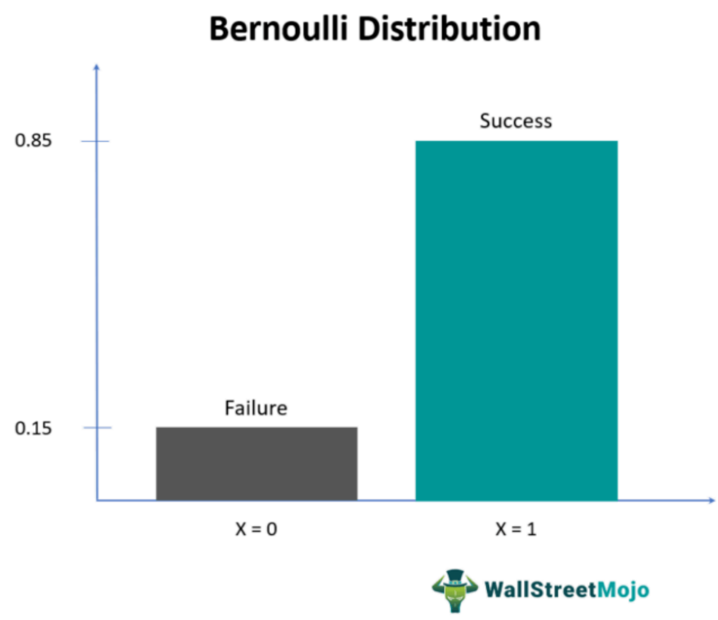
If you toss a coin:

- Success (Head) → X=1, p=0.5
- Failure (Tail) → X=0, 1−p=0.5

So,

P(X=1)=0.5,P(X=0)=0.5

🖼️ Visual Representation



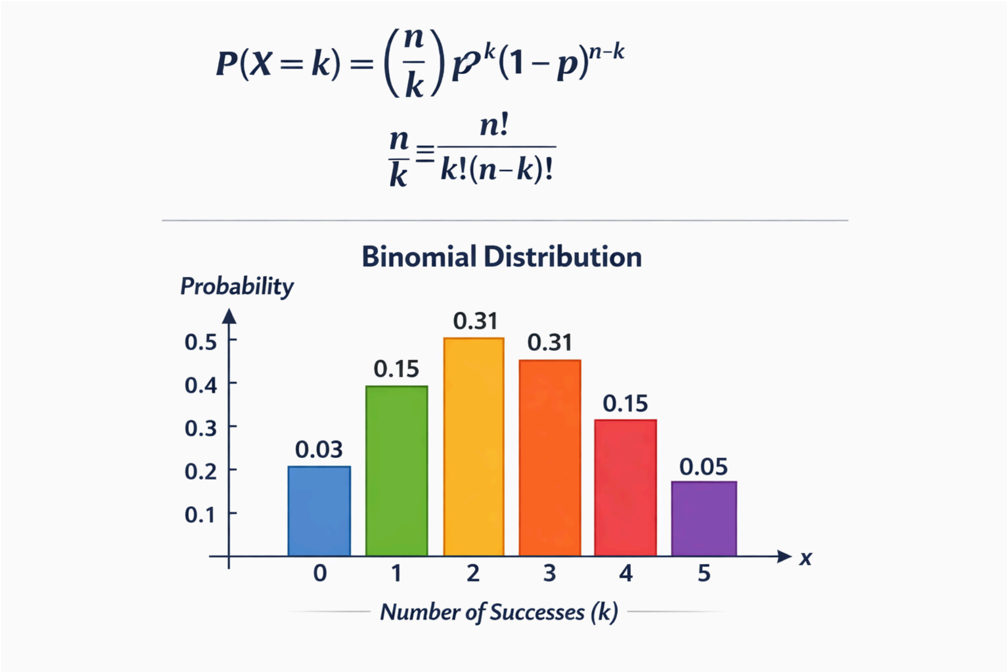
5: What is Binomial Distribution?

Answer:

The **Binomial Distribution** is a discrete probability distribution that models the number of **successes** in a fixed number of independent trials, where each trial has the same probability of success.

- **Parameters**:
  - n = number of trials
  - p = probability of success
  - $q=1-p$ = probability of failure
- **Mean**: $\mu=n\cdot p$
- **Variance**: $\sigma 2=n\cdot p\cdot(1-p)$

📐 Formula



📊Example

If you toss a coin **5 times**, and want the probability of getting **exactly 3 heads**:

n=5,p=0.5,k=3

P(X=3)=(53)(0.5)3(0.5)2
=10×0.125×0.25
=0.3125

So, the probability of getting **3 heads** is **0.3125**.

🖼️ Visual Representation

Above is a simple image showing the **Binomial Distribution formula** and probability bars for different success counts.

6: Explain Log-Normal Distribution

Answer:

The **Log-Normal Distribution** is a continuous probability distribution of a random variable whose **logarithm is normally distributed**.

It is used to model **positively skewed data** — where values cannot be negative and grow exponentially.

- **Characteristics**:
  - Always **positive**.

- ○ **Right-skewed** curve.
- ○ Common in **finance, biology, and economics**.

📐 Formula

Log-Normal Distribution

Mean: $\mu_X = e^{\mu + \sigma^2/2}$

Variance: $o_X^2 = (e^2 - 1)e^{2\mu + \sigma^2}$

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}}\, e^{-\frac{-(\ln x - \mu)^2}{2\sigma^2}},\ x > 0$$

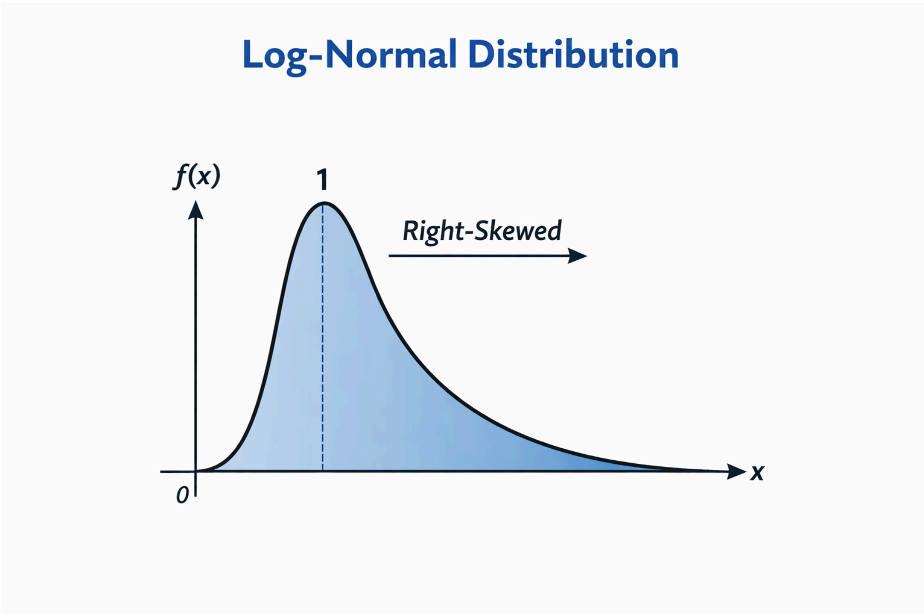
📊Example

If the **logarithm of income** follows a normal distribution, then the **income itself** follows a log-normal distribution.

This explains why most people earn moderate incomes, while a few earn extremely high ones.

🖼️ Visual Representation

Below is a simple image showing the **Log-Normal Distribution formula** and its **right-skewed curve**.



7: Explain Power Law Distribution

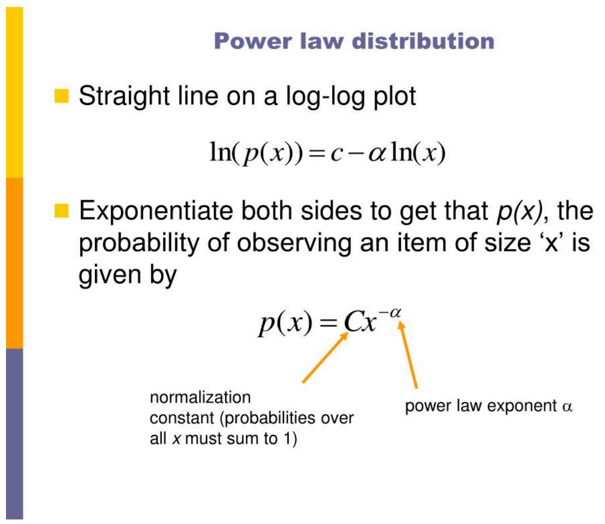
Answer:

The **Power Law Distribution** describes phenomena where **small occurrences are extremely common**, but **large occurrences are rare**.

It is often used to model **heavy-tailed data** — where a few values dominate the overall pattern.

- **Characteristics**:
    - Follows a **polynomial decay** pattern.
    - Common in **earthquakes, wealth, city populations, and internet traffic**.
    - Exhibits **scale invariance** — the same pattern repeats at different scales.

📐 Formula



where:

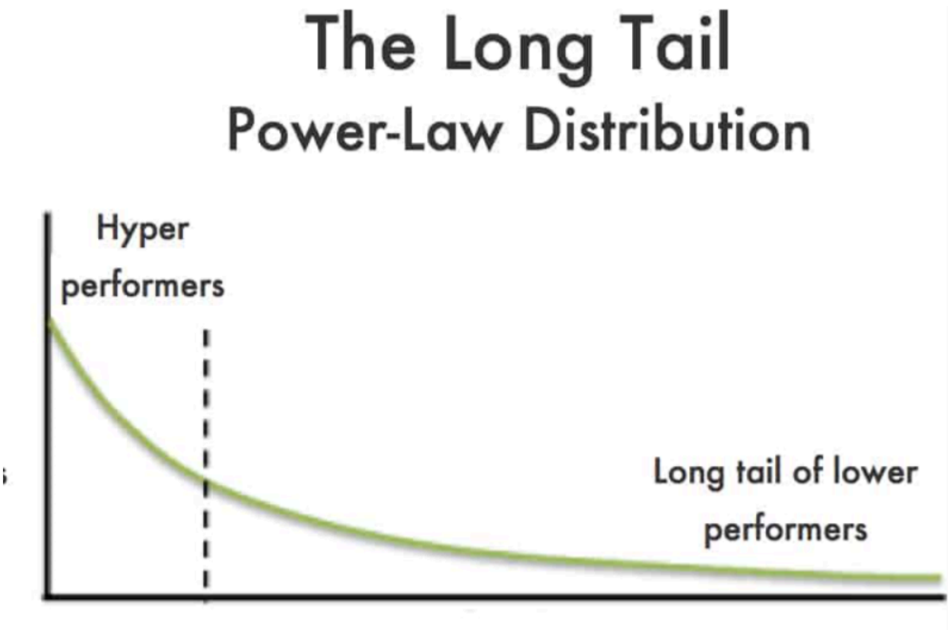
- C = normalization constant
- α = exponent (controls tail heaviness)
- xmin = minimum value for which the law holds

📊Example

- **Wealth Distribution**:
    - Many people have moderate wealth.
    - A few have extremely high wealth.
    - The probability of finding someone twice as rich decreases polynomially.

🖼️ Visual Representation

Below is a simple image showing the **Power Law Distribution formula** and its **heavy-tailed curve** — where the probability drops sharply as x increases.



8: Explain Box-Cox Transform

Answer:

The **Box-Cox Transform** is a statistical technique used to **stabilize variance** and make data more **normally distributed**.

It's especially useful when data is **skewed** or has **non-constant variance** in regression or time series analysis.

- **Purpose**: Converts non-normal data into a form closer to normal.
- **Parameter**: λ (lambda) — controls the transformation strength.
- **Applicable only** for **positive data values**.

📐 Formula

$$\begin{cases} y = \dfrac{x^{\lambda} - 1}{\lambda} \text{ where } \lambda \neq 0 \\ y = \ln x \text{ where } \lambda = 0 \end{cases}$$

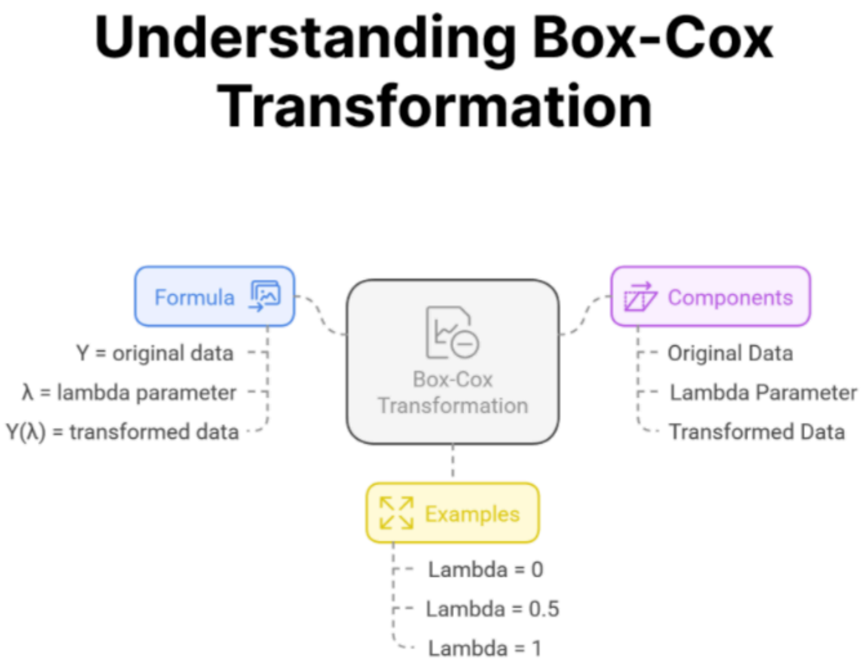
📊Example

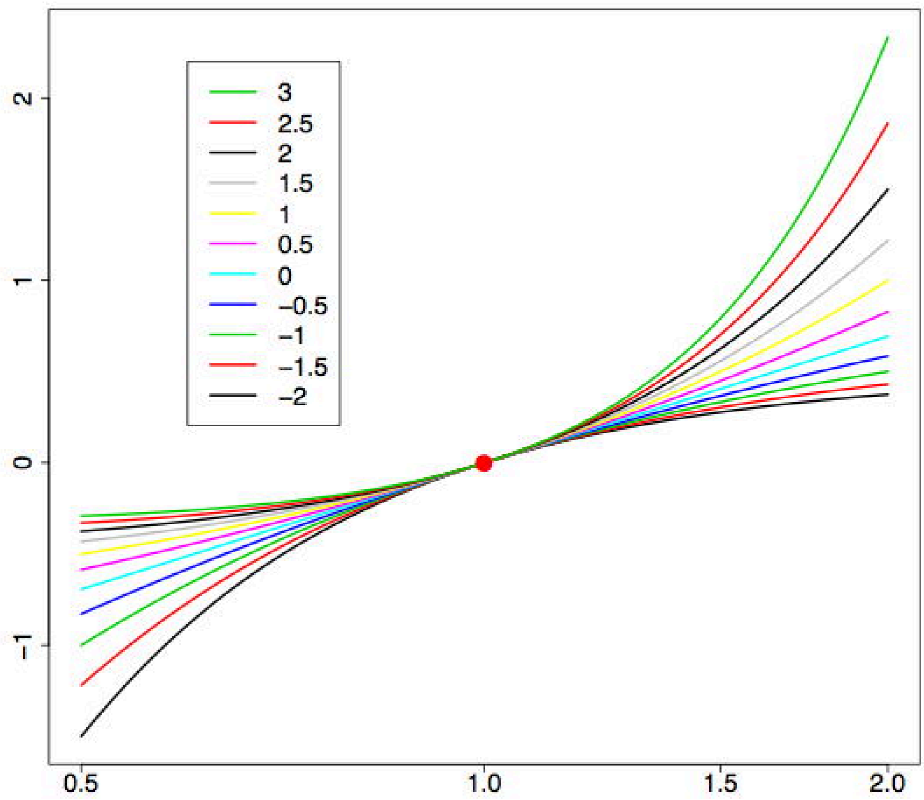
Suppose you have **sales data** that grows exponentially.

Applying a **Box-Cox transform** with λ=0.5 reduces skewness and makes the data more symmetric — helping regression models perform better.

🖼️ Visual Representation

Below is a simple image showing the **Box-Cox Distribution** and how the transformation changes the shape of data toward normality.





9: Explain Poisson Distribution

Answer:

The **Poisson Distribution** is a discrete probability distribution that models the number of times an event occurs in a fixed interval of time or space, given the events happen independently and at a constant average rate.

- **Parameter**: λ (lambda) = average rate of occurrence
- **Mean**: μ=λ
- **Variance**: σ2=λ
- **Applications**: Modeling arrivals (calls, emails), rare events (accidents, earthquakes).

📐 Formula

THE POISSON PROBABILITY FORMULA

$$P(X = K) = \frac{e^{--}\lambda^{K}}{K!}$$

Where: λ = average number of events per interval

K = actual number of events

e = Euler's number (≈ 2,718)

📊Example

If a call center receives **5 calls per hour on average** (λ=5):

- Probability of receiving exactly **3 calls in an hour**:

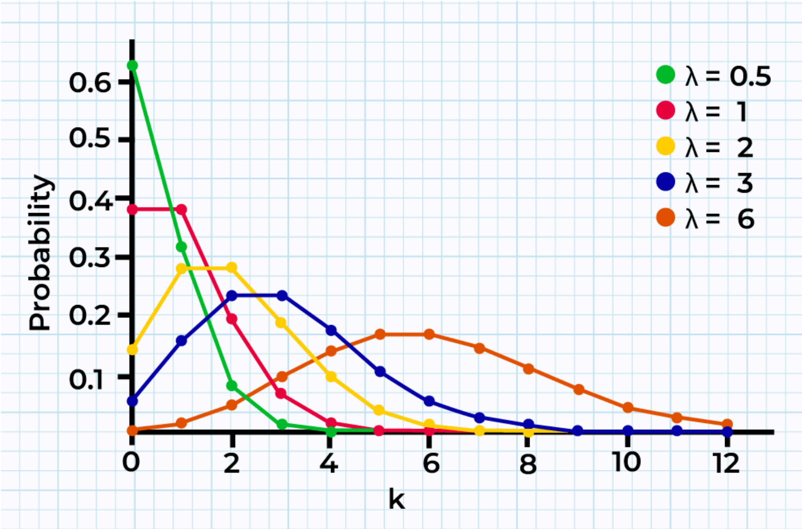
P(X=3)=e−5⋅533

!=0.1404

So, there's about a **14% chance** of 3 calls in that hour.

🖼️ Visual Representation

Below is a simple image showing the **Poisson Distribution formula** and its **probability bars** for different event counts.



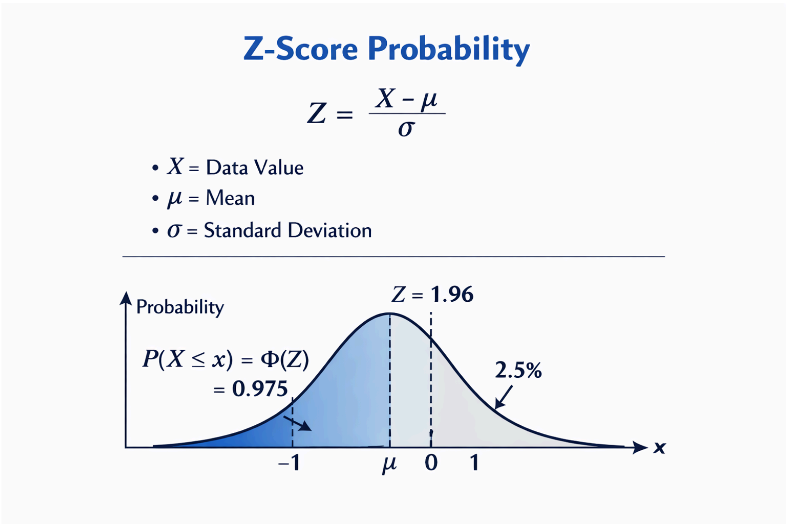
10: What is Z-score Probability?

Answer:

The **Z-score Probability** represents how far a data point is from the mean of a normal distribution, measured in **standard deviations**.

It helps find the **probability** or **area under the curve** corresponding to a given Z-value.

- **Formula**:



- **Interpretation**:
  - Z=0: value equals the mean.
  - Z=1: one standard deviation above the mean.
  - Z=−1: one standard deviation below the mean.

Probability:

- The probability is found using the **Cumulative Distribution Function (CDF)** of the normal distribution:

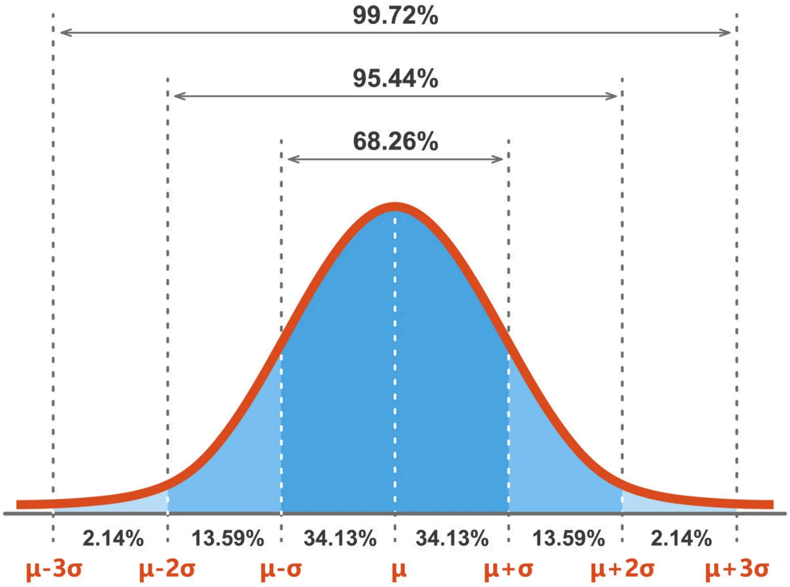
$$P(X \leq x) = \Phi(Z)$$

where Φ(Z) gives the area under the curve to the left of Z.

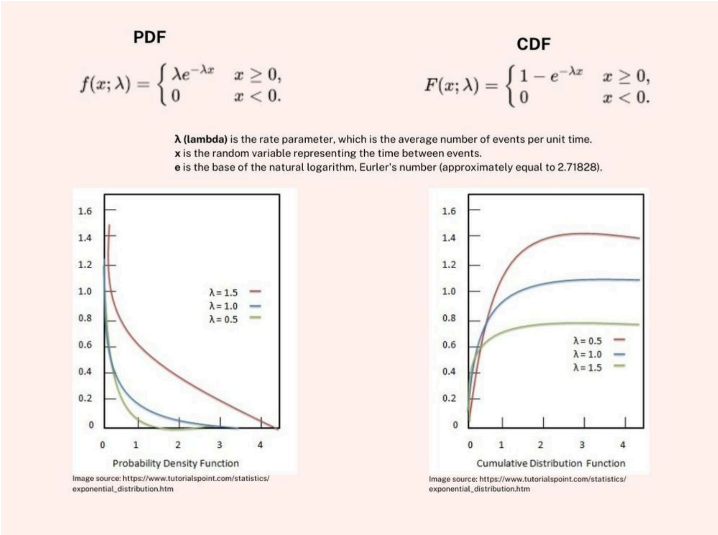
Example:

If Z=1.96, then P(X≤x)=0.975.

That means **97.5%** of values lie below 1.96 standard deviations from the mean.



11: Differentiate PDF and CDF



Feature	PDF	CDF
Full Form	Probability Density Function	Cumulative Distribution Function
Meaning	Shows likelihood at a specific value	Shows probability up to a value
Range	Can exceed 1 (height of curve)	Always between 0 and 1
Shape	Bell-shaped or other continuous curve	S-shaped increasing curve
Formula	f(x)=dF(x)dx	F(x)=∫−∞xf(t) dt
Use	To find probability density	To find cumulative probability

📊 Example:

Normal Distribution – PDF vs CDF

- **PDF:** Shows the **bell curve** — how data is spread around the mean.

  Example: For $\mu=0, \sigma=1$, the curve peaks at $x=0$.

- **CDF:** Shows the **probability** that a value is **less than or equal to** x.

  Example: For $Z=1.96$, $P(X \le x)=0.975$ → 97.5% of values lie below $x=1.96$.